Monetary Policy and Corporate Green Transition

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Abstract

This study investigates the impact of monetary policy on firms' green transition, as reflected in their carbon emissions and the elasticity of emissions to sales. We find that firms' carbon emissions increase following monetary policy tightening, suggesting that the reduction in green investment outweighs the decline in production. Moreover, consistent with the view that green investments are more interest rate-sensitive than brown investments, we also find that the elasticity of emissions to sales rises after monetary tightening. Corroborating these results, we show that firms reduce decarbonization projects, especially those with longer payback periods, resulting in a lower ratio of decarbonization investment to total investment. Cross-sectional analysis reveals that management control and stakeholder pressure, rather than financial constraints, largely explain the variations in the transmission of monetary policy. Overall, we contribute to the literature by documenting the disaggregate effect of monetary policy on corporate emissions and highlighting the heterogeneous responses across firms.

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I. INTRODUCTION

Global warming and extreme weather events have triggered worldwide recognition that accelerating the transition to net zero carbon emissions is essential. This transition requires firms to decarbonize their operations and switch from fossil fuels to renewable energy sources, a process that has attracted significant attention from policy makers and regulators worldwide. While carbon pricing instruments are the primary regulatory tool for reducing carbon emissions (Colmer et al. 2024), there is increasing recognition that monetary policy — one of the most significant government policies — can have unintended consequences in firms' carbon transition (ECB 2021; FEDS Notes 2021; Arseneau, Drexler, and Osada 2022). Since the main objective of monetary policy is to stimulate economic growth, previous research has largely focused on its effects on macroeconomic variables such as output, employment, and inflation. However, by influencing economic behavior, monetary policy also impacts firms' energy consumption and green investment. In this study, we examine whether and how monetary policy shapes firms' carbon transition, as reflected in both their carbon emissions and the elasticity of emissions to sales.

In the United States, the primary monetary policy tool is the federal funds rate (FFR), which is the overnight interest rate that banks charge each other. The FFR is typically set through announcements by the Federal Open Market Committee (FOMC) during its eight regularly scheduled meetings each year. While changes in the FFR are relatively small and short-term, they exert significant influence on the course of the real economy not only directly through the cost of capital but also indirectly through borrowers' balance sheets and banks' lending capacity (Bernanke and Gertler 1995; Mishkin 1995).

We first examine the impact of monetary policy on firms' carbon emissions because the level of emissions is linked to firms' carbon transition risk to net zero (Bolton and Kacperczyk 2021). A traditional view is that a tightening of monetary policy decreases aggregate consumer demand, industrial production, and corporate investment. The reduction in production should decrease energy consumption, which in turn lowers firms' carbon emissions. Conversely, an easing of monetary policy increases consumer demand and industrial production, leading to higher carbon emissions. We refer to this mechanism as the production effect of monetary policy on corporate carbon emissions.

However, a growing literature suggests that green investments are more sensitive to monetary policy than brown investments. First, renewable energy projects such as wind and solar power require relatively high upfront costs and risky payoff, making them particularly sensitive to interest rate changes (Eyraud, Clements, and Wane 2013; Egli, Steffen, and Schmidt 2018). Second, firms may undertake decarbonization projects due to political, social, and compliance reasons rather than NPV maximization (Achilles et al. 2024).[1] As interest rates rise, concerns about economic contraction and profitability pressure increase. In response, firms may prioritize cutting these investments. Consequently, a tightening of monetary policy would reduce green investments and the associated carbon savings, leading to an increase in carbon emissions, whereas an easing of monetary policy would lead to the opposite effect. We refer to this mechanism as the investment effect of monetary policy.

Taken together, these arguments imply that the net effect of monetary tightening on corporate carbon emissions is ultimately an empirical question, as it depends on the relative

[1] According to the CDP survey of global firms, firms reported over 12,500 decarbonization projects totaling approximately EUR 639 billion from 2015 to 2023 (Fuchs, Stroebel, and Terstegge 2024). The majority of these projects focus on low-carbon energy investments, building and process efficiency improvements, and initiatives to reduce transportation-related emissions.

strength of the production and investment effects. Anecdotal evidence, however, suggests that central bankers share the view that monetary tightening will slow down the green transition (Schnabel 2023). Following this view, our first hypothesis predicts that changes in monetary policy are positively associated with subsequent changes in firms' carbon emissions.

The level of emissions, however, is tightly linked to production and sales and does not accurately reflect the effect of changes in green investments on the carbon intensity of firm operations. Corporate emissions can rise or fall when firms expand or reduce operations using existing production methods, without reducing green or brown investments. To more directly capture the carbon impact of firms' changes in green versus brown investments, we examine the elasticity of carbon emissions to firm sales, defined as the percentage change in carbon emissions in response to a percentage change in sales revenue. This measure allows us to more accurately assess how fast firms adjust the carbon intensity of their operations in response to monetary policy, holding the output level constant. If a tightening of monetary policy results in a greater reduction in a firm's green investments relative to brown investments, we expect its elasticity of emissions to sales to increase. Consequently, our second hypothesis predicts that changes in monetary policy are positively associated with subsequent changes in the elasticity of firms' carbon emissions to sales.

We test our hypotheses using a sample of 4,582 firm-year observations for 603 non-financial firms that voluntarily disclose Scope 1 (direct) carbon emissions from 2005 to 2021.[2] We focus on firm-reported carbon emissions rather than vendor-estimated emissions because the firm-reported data are arguably less noisy and more accurately reflect firms' emission

[2] The GHG Protocol classifies greenhouse gas (GHG) emissions into three scopes based on the sources of emissions. Scope 1 emissions are the direct emissions that originate from sources owned or controlled by a firm. Scope 2 emissions are the indirect emissions associated with the generation of electricity, heat, or steam that a firm purchases and consumes. Scope 3 emissions are the other indirect emissions that occur in a firm's value chains.

performance (Aswani, Raghunandan, and Rajgopal 2024). Because most large firms voluntarily report their emissions due to pressure from stakeholders, our sample is biased toward large firms. However, it includes diverse representations from various industries and covers companies that face significant public scrutiny and are crucial to the transition to net zero.

Since firms disclose carbon emissions annually, we align our measurement of monetary policy changes to the same frequency. We focus on one-year-ahead emissions outcomes in our main analysis because it takes time for the monetary policy to affect real investments (Bernanke and Gertler 1995). We use two complementary proxies for monetary policy: changes in the FFR and shocks to the FFR. While changes in the FFR are straightforward to measure, they may reflect other macroeconomic factors and the central bank's assessment of the current and future state of the economy. To address this concern, we also employ monetary policy shocks following Jarociński and Karadi (2020), who identify these shocks based on the high-frequency changes in interest rate futures and stock prices around FOMC announcements. This approach, widely used in the literature, captures the unanticipated component of monetary policy and helps isolate it from both concurrent macroeconomic news and the Fed's information set.

We test our first hypothesis by regressing changes in the natural log of a firm's Scope 1 emissions (i.e., direct emissions from owned or controlled sources) on lagged monetary policy changes. We control for a variety of macroeconomic and firm characteristics and include firm fixed effects. Consistent with our prediction, we find that both FFR changes and FFR shocks are positively associated with subsequent changes in the log of Scope 1 emissions. The results are economically significant. On average, a 1% increase in the FFR leads to a 1.5% increase in a firm's Scope 1 emissions.

To test our second hypothesis, we augment the regression model with an interaction term between monetary policy changes and changes in the natural log of a firm's sales. Because this specification focuses on the interaction term, it allows us to add year fix effects to account for time-varying unobserved macroeconomic conditions. As predicted, we find that both FFR changes and FFR shocks are positively associated with subsequent changes in the elasticity of Scope 1 emissions to firm sales. In terms of economic significance, a one-standard deviation increase in the FFR amplifies the elasticity of Scope 1 emissions to firm sales by 19.9%.

Our results are robust to using alternative samples (e.g., firms with carbon assurance), alternative measures of FFR shocks, and alternative cluster schemes. Collectively, these findings suggest that following monetary tightening, managers are more inclined to reduce green investment compared to non-green investment, leading to higher carbon emissions and more carbon-intensive production.

To bolster our inferences, we explore the effect of monetary policy on firms' decisions to invest in decarbonization projects. We take advantage of granular project-level data from the CDP climate change corporate response dataset, the largest and most comprehensive source of corporate climate action (Achilles et al. 2024; Fuchs et al. 2024). Among our sample firms, most of the decarbonization investments relate to building and process efficiency. We find that both the annual amount of decarbonization investments implemented and the payback period for these projects decrease following a rise in interest rates. Additionally, the ratio of decarbonization investment to total investment also declines. These findings support the view that following monetary tightening, firms defer capital-intensive, long-term green initiatives to minimize exposure to costly financing.

Next, we perform several cross-sectional analyses to explore the heterogeneity in the impact of monetary policy on firms' carbon transition. We focus on firms' internal carbon management control systems and external stakeholder pressure. These governance mechanisms, well documented for their influence on firms' environmental externalities (Ionnou, Li, and Serafeim 2017; Dyck et al. 2019; Krueger et al. 2020), reflect firms' carbon commitments and investors' awareness of climate change risk. If monetary tightening leads firms to reduce green investments due to concerns of financial performance, we expect this tendency to be mitigated by firms' carbon management control and stakeholder pressure. Conversely, if monetary easing motivates firms to increase green investments, we expect the changes to be muted among firms with strong internal carbon management and external stakeholders because these firms already make green investments to meet the carbon target or stakeholder expectations.

We measure a firm's internal carbon management systems based on whether it sets monetary incentives and targets for carbon emissions. We capture external stakeholder pressure based on whether a firm is located in a democratic-leaning county (Di Giuli and Kostovetsky 2014) or has high ownership by signatories of the United Nations-supported Principles for Responsible Investment (UN PRI). Consistent with our expectations, the positive association between monetary policy and firms' elasticity of emissions to sales is weaker among firms with stronger carbon management control systems and under greater pressure from environmentally conscious stakeholders. These findings suggest that while firms' decarbonization investments are relatively sensitive to changes in monetary policy, strong corporate commitment and investor awareness of climate risk help mitigate this sensitivity and enhance firms' carbon accountability.

An alternative explanation is that firms are financially constrained from making green investments following rising interest rates. This is unlikely to explain our results because the firms in our sample are large and mature, and thus generally do not lack the ability to access external funding. Additionally, while prior research finds that the impact of monetary policy on *total investment* is stronger among financially constrained firms (Gertler and Gilchrist 1994; Kashyap, Lamount, and Stein 1994), this result may not be applicable in our setting because firms choose *decarbonization investments* based not only on the cost of capital, but also on political, social, and reputational considerations. Moreover, if firms with more financial constraints already invest less in green investments (Xu and Kim 2022), these firms will have less ability to cut green investments relative to other firms in response to rising interest rates. Additional analyses confirm this expectation and find insignificant differences in the association between monetary policy and elasticity of emissions to sales across the subsamples conditional on three proxies of financial constraints: firm age, size, and a composite index based on Hadlock and Pierce (2010).

Our study makes several contributions to the literature. First, we bridge two distinct but increasingly relevant strands of research: the rapidly growing literature on climate finance (e.g., Hong, Karolyi, and Scheinkman 2020; Bolton and Kacperczyk. 2021; Giglio, Kelly, and Stroebel 2021) and the established body of work examining the effects of monetary policy on the real economy (Gertler and Gilchrist 1994; Bernanke and Gertler 1995; Ottonello and Winberry 2020; Gurkaynak, Karasoy-Can, and Lee 2022). While studies have examined the impact of monetary policy on environmental sustainability, they primarily focus on aggregate country-level carbon emissions and find mixed results (e.g., Qingauan et al. 2020; Pradeep 2022; Attílio, Faria, and

Rodrigues 2023; Anastasiou et al. 2024).[3] We offer the first evidence on firm-level disaggregate effects.

Second, we complement studies that examine the impact of management control on firms' green performance. A substantial body of research examines the roles of monetary incentives and target settings in managerial performance, yielding mixed results (Bonner and Sprinkle 2002; Indejikian, Matejka, and Schloetzer 2014). Recent studies document the growing adoption of carbon metrics in executive compensation and managerial targets, and suggest that these tools help enhance environmental performance (Ionnou et al. 2017; Cohen et al. 2023; de Haas et al. 2025). We differ from these studies by showing that carbon management tools mitigate the negative impact of monetary shocks on firms' environmental externalities.[4]

Finally, we add to the growing literature that examines the heterogeneous firm-level effect of monetary policy (Armstrong et al. 2019; Gallo and Kothari 2019; Leahy and Thapar 2022).[5] Our study suggests that the firm-level heterogeneity in the transmission of monetary policy on corporate carbon emissions is largely driven by firms' internal and external carbon governance, rather than financial constraints. These findings enrich the ongoing debate about the evolving role of central banks in addressing climate-related risks. Traditionally, central banks have focused on prices and financial stability. As climate risks increasingly affect economic and financial systems, they are increasingly concerned about the impact of monetary decisions on

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[3] Qingauan et al. (2020) and Attílio et al. (2023) find a negative relation between interest rates and carbon emissions in Asian economies and the U.S. whereas Pradeep (2022) finds a positive relation in India. The results are difficult to compare because of institutional differences and endogenous nature of country-level data.

[4] Our work also extends research examining the influence of various stakeholders on corporate social responsibility (CSR) practices. For instance, Krueger, Sautner, and Starks (2020) find that institutional investors actively engage with firm management to mitigate climate risk. Dyck et al. (2019) show that institutional investors play a key role in driving CSR performance globally. Di Giuli and Kostovetsky (2014) provide evidence that a firm's environmental performance is influenced by the political leaning of the state where its headquarters are located.

[5] As noted by the former Fed chair Janet Yellen, "Studying monetary models with heterogeneous agents more closely could help us shed new light on these aspects of the monetary transmission mechanism." (Yellen 2016).

environmental sustainability (Arseneau et al. 2022).[6] Thus, our study provides timely implications for central banks' consideration of incorporating sustainability factors into their monetary frameworks.

II. INSTITUTIONAL BACKGROUND AND HYPOTHESIS DEVELOPMENT

Monetary Policy and Climate Change

Monetary policy encompasses the actions taken by the central bank to manage the money supply or interest rates to achieve macroeconomic targets such as output, employment, and inflation. In the United States, the primary policy instrument is the FFR target. The FFR is typically set during the FOMC meetings, which occur eight times a year, approximately every six weeks. While the changes in the FFR are relatively small and short-term, a large body of research shows that they have far-reaching consequences in real economies beyond the narrow interest rate channel (Bernanke and Gertler 1995; Mishkin 1995).

To explain the transmission of the FFR, Bernanke and Gertler (1995) discuss several channels that they collectively refer to as "the credit channel of monetary policy transmission." In addition to the traditional cost-of-capital channel, they propose the balance sheet channel and the bank lending channel. The balance sheet channel addresses the potential impact of changes in the FFR on borrowers' balance sheets. When interest rates rise, the financial positions of households and firms deteriorate due to increased interest expenses and reduced collateral value. The weakened balance sheet discourages borrowers from taking additional debt, thereby reducing consumption and investment. The bank lending channel focuses more narrowly on the supply of bank loans. Under this channel, rising FFR drains bank reserves and deposits through

[6] Also, see "ECB presents action plan to include climate change considerations in its monetary policy strategy" in the ECB's press release on July 8, 2021.

the open market sale by the Fed. The reduced access to loanable funds, in turn, limits the supply of bank loans.

Given the growing recognition of climate change as a risk to financial stability, central banks around the world have become increasingly attentive to the implications of monetary policy for the green transition. For example, a release by the U.S. Federal Reserve notes: “The Fed is concerned that climate change-related financial risks could impact both individual financial institutions and the broader financial system, potentially affecting financial stability (FEDS Notes 2021). Similarly, Christine Lagarde, President of the European Central Bank (ECB), emphasized in a hearing before the European Parliament’s Committee on Economic and Monetary Affairs that the ECB's climate action plan “sets out an ambitious timeline and outlines a wide range of actions,…ultimately aiming to consistently integrate climate change considerations into all aspects of the ECB’s monetary policy” (ECB 2021).

Hypothesis Development

The extant literature suggests that monetary tightening reduces overall consumer demand, industrial production, and corporate investments. These changes can affect firms’ carbon emissions in opposite directions, with the net effect unclear. Based on the *production effect* of monetary policy transmission, higher interest rates reduce firms’ total output and energy consumption, thereby lowering direct carbon emissions. Under this mechanism, monetary tightening is expected to decrease firms’ carbon emissions, while monetary easing is likely to increase them.

In contrast, the *investment effect* of monetary transmission suggests that monetary tightening may raise firms' carbon emissions. This is because green investments[7] tend to be more sensitive to interest rate changes than brown investments. First, higher interest rates increase the cost of credit, which diminishes the present value of the investment. Green investments, such as renewable technologies embedded in new machinery and vehicles, are particularly exposed to rising borrowing costs due to their capital-intensive nature (Egli et al. 2018). An analysis by Wood Mackenzie, an energy consulting firm, reveals that a 2-percentage point rise in the risk-free interest rate can increase the levelized cost of electricity (LCOE) for renewables by as much as 20%. In contrast, the LCOE for a combined-cycle gas turbine plant increases by only 11% (Martin et al. 2024).[8]

Consistent with this argument, a speech by Isabel Schnabel, Member of the Executive Board of the ECB, on January 10, 2023, states:

> *The green transition will fundamentally transform our societies. Protecting our planet requires unprecedented large-scale investments in technical innovations and renewable energies to bring our economies on a path towards net zero greenhouse gas emissions.*
>
> …
>
> *These developments now risk being reversed by the marked rise in global interest rates over the past year. Since fossil fuel-based power plants have comparably low upfront costs, a persistent rise in the cost of capital may discourage efforts to decarbonise our economies rapidly.* (Schnabel 2023)

Second, firms' environmental performance is an outcome of various factors including managerial incentives, stakeholder expectations, reputational concerns, and regulatory pressure. While their environmental spending is associated with social returns, research suggests that firms tend to reduce their environmental spending when facing financial performance pressure (Liu et

[7] Survey evidence indicates that corporate green investments mostly involve process and equipment efficiency improvements—such as energy management, machinery upgrades, and waste minimization—rather than radical green patents or breakthrough innovations (Kalantzis et al. 2024).

[8] For an example of firms' responses to rising interest rates on green investments, see "GE Vernova boss keeps search for offshore wind turbine orders on ice" (*Financial Times*, November 12, 2024).

al. 2021). Because monetary tightening increases financial performance pressure by raising external financing costs, managers are likely to prioritize reduction of the decarbonization projects to boost short-term profitability.

While the overall effect of monetary policy on firms' carbon emissions is an empirical question, we propose our hypothesis based on the investment effect of monetary transmission given the concern of central bankers. Our first hypothesis (in alternative form) follows:

> *Hypothesis 1:* Changes in monetary policy are positively associated with subsequent changes in firms' carbon emissions.

To help us better understand the carbon impact of firms' changes in green versus brown investments, we also examine the elasticity of carbon emissions to firm sales, which is a more direct proxy for the greenness of firm operations. Although firms' absolute emissions are directly linked to economy-wide emissions and the goal of net zero, they do not accurately reflect the efforts toward decarbonization of their operations (Aswani et al. 2024; Zhang 2025). Firms' emissions will rise (or fall) if they expand (or reduce) operations using current production methods instead of actively cutting back (or investing) in decarbonization projects. If a tightening of monetary policy leads to a greater reduction in a firm's green investments compared to its brown investments, we expect the elasticity of emissions to sales to increase. Our second hypothesis (in alternative form) follows:

> *Hypothesis 2:* Changes in monetary policy are positively associated with subsequent changes in the elasticity of firms' carbon emissions to sales.

It is worth noting that to reduce climate-related risks and reputational damages, many firms have adopted internal carbon targets or provided monetary incentives for managers to reduce emissions (Ioannou, Li, and Serafeim 2017; Cohen et al. 2023).[9] Additionally, firms' emissions performance is under great scrutiny from their stakeholders (e.g., governments, investors, customers, and employees). Numerous institutional investors have adopted an investment approach that incorporates sustainability factors in the selection and management of their portfolios. To the extent that the internal and external carbon governance (i.e., management control tools and stakeholder pressure) incentivize managers to reduce carbon emissions, firms may not cut back on investments in clean technologies and environmental protection even in the face of rising interest rates.

III. SAMPLE, DATA, AND RESEARCH DESIGN

Sample

Our primary carbon emissions database is the S&P Trucost, a widely used commercial vendor of corporate carbon data (Bolton and Kacperczyk 2021). Trucost provides corporate carbon emissions data based on firm disclosure (from various public sources, such as annual financial reports, sustainability reports, and CDP surveys) or estimates using its environmentally extended input/output profiling model when a covered firm does not disclose carbon information. We focus on firm-reported Scope 1 carbon emissions to avoid potential noises associated with third-party estimated emissions (Aswani et al. 2024). This selection criterion enables a more precise examination of the environmental impacts of changes in firm production or operations in response to shifts in monetary policy. Our initial sample consists of 6,923 firm-years for 1,119 unique US firms that disclosed Scope 1 carbon emissions from 2005 to 2021. We then merge

[9] For example, the Coca Cola company states in its 2022 Business and Sustainability Report "we've set a science-based target to reduce absolute greenhouse gas emissions by 25% by 2030, against a 2015 baseline. As of 2022, we have reduced our emissions by 7% against this baseline."

Trucost with Compustat to obtain firm-level financial data, resulting in 6,820 firm-years for 1,095 unique firms. We gather macroeconomic variables, including the federal funds rate (FFR), the consumer price index (CPI), industrial production, real GDP growth, and employment rate from the Federal Reserve Economic Data (FRED). After excluding financial firms (SIC codes 6000–6999) and observations with missing control variables, our final sample comprises 4,582 firm-year observations for 603 unique firms.

Because we focus on voluntary disclosers of carbon emissions, our sample is biased toward larger companies. However, these firms represent the most relevant population for addressing our research question due to their significant contribution to carbon emissions. Table 1 Panel A outlines the sample selection process. Panel B reports the sample distribution by industry. It shows that the capital goods, materials, and utilities sectors have the largest number of observations.

We obtain data on firms' decarbonization initiatives and green management control tools from the CDP climate change corporate response dataset. The CDP is an international nonprofit organization that conducts annual surveys of the largest companies in each country on behalf of its institutional investor signatories. The CDP surveys aim to assess the risks and opportunities associated with climate change, providing investors with information to evaluate firms' climate-related risks. They also help firms develop the capability to generate and disclose comparable and relevant climate data to their stakeholders, which is generally unavailable in regulatory filings.

Variables

Measuring Corporate Carbon Emissions and Decarbonization Investments

We measure a firm's carbon emissions using the natural log of its disclosed Scope 1 carbon emissions from Trucost. We measure firms' decarbonization investments using the project-level data on carbon reduction initiatives from the CDP survey. Specifically, the survey asks, "Did you have emissions reduction initiatives that were active within the reporting year?" If the answer is "Yes," firms are requested to provide details on the initiatives implemented in the reporting year, including the type of initiative, the estimated annual carbon savings, the total amount of investment required, the payback period, and the estimated lifetime of the initiative.

Appendix A provides excerpts from Cargill's 2022 CDP Climate Change Questionnaire (for the reporting year of 2021). Panel I shows that among the implemented projects, an energy efficiency initiative in production processes is estimated to yield annual carbon savings of 3,147 metric tonnes, requiring an investment of $601,504 and an expected payback period of one to three years.

Measuring Monetary Policy Changes

We construct two measures of monetary policy changes. The first one is the changes in the federal funds rate. This rate is the overnight interest rate banks charge each other, which is a critical tool for the US Federal Reserve to set monetary policy.[10] Numerous studies suggest that this policy instrument significantly affects the financial markets and the real economy through the credit channel (Bernanke and Gertler 1995; Armstrong, Glaeser, and Kepler 2019). To align with the annual frequency of reported emissions data, we calculate the annual change in the rate.

Our second measure is a funds-rate shock based on the monetary policy shock identified by Jarociński and Karadi (2020). This measure is derived from high-frequency movements in interest rate futures with maturities ranging from one month to one year, observed within a 30-minute window around FOMC announcements. Jarociński and Karadi (2020) further refine the

[10] An exception during our sample period is from December 2008 to December 2015 when the federal funds rate is near zero, because the Federal Reserve cannot move the rate below 0% (i.e., the zero lower bound).

identification by employing a vector autoregression (VAR) model with sign restrictions to disentangle monetary policy shocks from information shocks. Specifically, the monetary policy shock is based on the negative high-frequency co-movement between changes in interest rates and stock prices around FOMC announcements. The underlying logic is that a pure monetary tightening raises the discount rate and reduces expected dividends, thereby lowering stock market valuations. In contrast, a complementary information shock, which reveals positive news about the economic outlook, raises stock market valuations by increasing expected dividends. By disentangling these effects, the monetary policy shock reduces the bias of conflating monetary policy actions with the central bank's assessment of the economic outlook, making it more suitable for identifying the effects of monetary policy. We aggregate the monetary policy shock to the annual level to match the frequency of emissions data.

Descriptive Statistics

We winsorzie all continuous variables at the top and bottom 1% level to mitigate the influence of outliers. Appendix B provides variable definitions and data sources. Table 2 Panel A presents summary statistics for the variables. It shows the average change in the log of Scope 1 emissions is -0.011, corresponding to an annual decline of approximately 1.1% in Scope 1 emissions. The average change in the federal fund rates is -0.149 percentage points during our sample period. Panel B reports the Pearson's correlation coefficients. We find that changes in emissions (*ΔLog(Scope 1)*) are positively associated with changes in the federal funds rate (*ΔFFR*) and with federal funds rate shocks (*FFR Shock*).

Research Design

We test our first hypothesis by estimating the following regression:

$$\Delta Log(Scope1)_{it} = \beta_1 \Delta MP_{t-1} + \gamma_1 Controls_{t-1}^{m} + \gamma_2 Controls_t^{f} + \mu_i + \varepsilon_{it} \tag{1}$$

where i denotes voluntary carbon disclosure firms and t denotes years. $\Delta Log(Scope1)_{it}$ is the change in the natural log of Scope 1 carbon emissions. ΔMP_{t-1} captures monetary policy changes, measured as changes in the federal fund rate (ΔFFR) or as funds-rate shocks (*FFR Shock*). We use the lagged monetary policy measure because interest rate decisions take time to affect investment decisions and influence emission levels.[11] This time lag also mitigates the concern of reverse causality. Our first hypothesis predicts β_1 to be positive.

Following prior studies (Döttling and Ratnovski 2023; Ma and Zimmerman 2023), we control for several lagged macroeconomic variables: the industrial production index (*Log industrial production)*, consumer price index (*log CPI*, employment rate), real GDP growth (*ΔGDP)*, and excess bond premium (*Excess Bond Premium*). The excess bond premium, from Gilchrist and Zakrajšek (2012), captures time-varying risk premia in credit markets and serves as a proxy for financial market sentiment and uncertainty. We also include an indicator for the period from 2009 to 2015 to account for the time when the FFR is near zero (*D_Zero Bound*). In addition, we control for various firm characteristics, including cash holdings (*Cashhold*), cash flows (*Cashflow*), return on asset (*ROA*), leverage (*Leverage*), firm size (*Size*), asset tangibility (*Tangibility*), market valuation (*Tobin's Q*), and changes in the log of firm sales ($\Delta Log(Sale)$). We include firm fixed effects (μ_i) to account for time-invariant, unobserved firm heterogeneity. We adjust standard errors using firm and year two-way clusters in all regressions.

To test our second hypothesis, we expand Equation (1) with an interaction term between monetary policy changes and changes in the log of firm sales. Our regression model is specified as follows:

[11] Bernanke and Gertler (1995) find that the decline in fixed investments in response to a monetary tightening shock occurs between six to 24 months after the shock. This decline reverses after 24 months, likely because the interest rate is virtually back to trend eight to nine months after the monetary tightening.

$$\Delta Log(Scope1)_{it} = \beta_1 \Delta MP_{t-1} \times \Delta Sale_{it} + \beta_2 \Delta Sale_{it} + \gamma_1 Control_t^f + \mu_i + \eta_t + \varepsilon_{it} \quad (2)$$

where i denotes voluntary carbon disclosure firms and t denotes years. Our variable of interest is the coefficient on the interaction term, β_1, which captures the impact of monetary policy changes on the elasticity of carbon emissions to sales (i.e., the percentage change in carbon emissions in response to a percentage change in sales revenue). Our second hypothesis predicts β_1 to be positive.

We control the same time-varying firm characteristics and firm fixed effects as in Equation (1). Since the main variable of interest is the coefficient on the interaction term, we replace lagged macroeconomic controls with year fixed effects (η_t) to control for time-varying unobserved macroeconomic conditions.

IV. EMPIRICAL ANALYSIS

Test of Hypothesis 1: Effect of Monetary Policy on Corporate Carbon Emissions

Table 3 presents the results of the average effect of monetary policy changes on the subsequent changes in corporate carbon emissions. Columns (1)-(3) report results using the changes in the FFR as the independent variable and Columns (4)-(6) report results using FFR shocks. Columns (1) and (4) include our variable of interest, lagged macroeconomic control variables, and firm fixed effects. Columns (2) and (5) add firm-level controls, excluding the change in the log of sales. Columns (3) and (6) further control for the change in the log of sales.

Consistent with our first hypothesis, we find that the coefficients on ΔFFR and *FFR Shock* are significantly positive, at $p \leq 0.10$ in all models. The results are also economically meaningful. For example, Column (3) shows that a 1% increase in the FFR is, on average, associated with a

1.5% increase in a firm's Scope 1 carbon emissions. Overall, these findings suggest that firms' carbon emissions increase (decrease) following a monetary tightening (easing).

Test of Hypothesis 2: Effect of Monetary Policy on the Elasticity of Carbon Emissions to Sales

Table 4 presents the results of the average effect of monetary policy changes on the elasticity of carbon emissions to sales. Columns (1)-(2) report results using FFR changes and Columns (3)-(4) report results using FFR shocks. Columns (1) and (3) include our variable of interest and firm and year fixed effects. Columns (2) and (4) add firm-level controls. Consistent with our second hypothesis, the coefficients on the interaction terms $\Delta FFR \times \Delta Log(Sale)$ and $FFR\ Shock \times \Delta Log(Sale)$ are significantly positive, at $p \leq 0.10$ in all models. The result is also economically significant. For example, Column (2) indicates that a one standard deviation increase in the FFR (1.104 percentage points) leads to an approximate 19.9% increase in carbon emissions elasticity to sales.[12]

Taken together, the results in Table 4 provide support that the carbon intensity of firms' operations increases following monetary tightening. The robustness of the results using both measures of monetary changes reinforces that the documented effect is driven by monetary policy shocks, rather than the communication of the central bank's views about the state of the economy (Jarocinski and Karadi 2020).

Mechanisms through which Monetary Policy Affects Corporate Carbon Emissions

To further substantiate our inferences, we analyze how monetary policy affects firms' decarbonization investments using granular data from the CDP, which provides detailed project-level information on all carbon reduction projects that each firm implements each year. We

[12] 19.9%=0.071×1.104/0.394, where 0.071 is the coefficient on the interaction term, 1.104 is the standard deviation in the federal funds rate, and 0.394 is the coefficient on $\Delta Log(Sale)$.

restrict our analysis to the projects that have nonmissing monetary investments or payback periods and are related to Scope 1 emissions reductions. The sample for this analysis consists of 3,956 projects for 352 firms.

Table 5 Panel A summarizes the implemented decarbonization projects by the category and type, as classified by the CDP. Among our sample firms, the two most common carbon reduction initiatives are building efficiency (43.8%) and process efficiency (36.2%). Examples of building efficiency initiatives include upgrading Building insulation, Heating, ventilation, and air conditioning (HVAC), and building Energy Management Systems (BEMS). Examples of process efficiency include process optimization, process material efficiency, and equipment replacement.

Table 5 Panel B presents the summary statistics on the amount of decarbonization investments and the payback period. To assess the relative spending on decarbonization investments to total investment, the bottom row presents the statistics on the relative decarbonization investment, measured as natural log of project-level decarbonization investments divided by the firm-level sum of capital expenditure, SG&A and R&D.

Next, we examine the effect of monetary policy on the amount of decarbonization projects implemented each year and the payback period of these projects. Table 5 Panel C presents the results. Columns (1) and (2) indicate that the amount of decarbonization projects decreases following monetary tightening, suggesting that firms scale back their green investments in response to higher borrowing costs. Columns (3) and (4) reveal that firms significantly shorten the payback period of their decarbonization projects, implying a shift toward investments with quicker returns. Consistent with green investments being more sensitive to non-green investments, Columns (5) and (6) show that the ratio of decarbonization investment to total investment declines following monetary tightening. [13]

Collectively, the results in Table 5 suggest that monetary tightening reduces firms' capital allocation to decarbonization projects and shifts their preferences toward those with shorter payback periods. This change in corporate investment behavior slows the pace of economy-wide decarbonization, as fewer resources are directed toward large long-term projects necessary for achieving net-zero emissions. These findings reinforce our inferences, illustrating that monetary policy shapes firms' climate strategies and decarbonization efforts.

V. ADDITIONAL ANALYSES

In this section, we explore the heterogeneity in the impact of monetary policy on firms' carbon transition. We conduct analyses conditional on firms' internal carbon management control systems and external stakeholder pressure, followed by analyses conditional on firms' financial constraints. We use the models testing our second hypothesis to draw conclusions because they better capture firms' decarbonization efforts by controlling for output levels and incorporating firm and year fixed effects.

Analyses Conditional on Management Control Systems

If monetary tightening reduces firms' decarbonization effort due to increased financing costs and performance pressure, we expect this effect to be weaker among companies with a strong commitment to climate-related performance. We test this prediction by examining

[13] Our results are robust when using the ratio of the aggregated firm-level decarbonization investment to firm-level total investment.

whether firms with stronger carbon management control systems exhibit a weaker response to monetary policy changes in terms of their emissions-to-sales elasticity.

Following prior research (Lazear 1986; Bonner and Sprinkle 2002; Ioannou et al. 2017), we measure the strength of a firm's carbon management control system using monetary incentives and target setting. We obtain the data from the CDP annual surveys, which ask whether firms provide monetary incentives to manage climate-related goals and adopt carbon targets for the reporting year. Using Cargill as an example, Appendix A Panels II and III show that the company provides monetary rewards and adopts both absolute and intensity targets in 2021.

We perform the analysis by splitting the sample based on whether a firm-year adopts a specific carbon management control tool. Table 6 Panel A reports the results conditional on whether a firm-year offers monetary incentives for managing climate-related issues.[14] Columns (1) and (2) show the results using changes in the FFR. We find that for firm-years without the monetary incentives, the coefficient on the interaction term $\Delta FFR \times \Delta Log(Sale)$ is 0.245, which is more than three times larger than the full-sample estimate of 0.071 in Column (2) of Table 4. In contrast, for firm-years with the monetary incentives, this coefficient is much smaller in magnitude and statistically insignificant. The difference across the subsamples is also significant. Columns (3) and (4) show that the results are similar when comparing the coefficient on the interaction term $FFR\ Shock \times \Delta Log(Sale)$ across firm-years with and without the monetary incentives. These findings suggest that climate performance-based compensation mitigates the effect of monetary policy on firms' carbon transition.

Table 6 Panel B reports the results conditional on whether a firm-year adopts any carbon targets (i.e., absolute targets or intensity targets). We find that firm-years without a carbon target

[14] The total number of observations in Table 6 is smaller than the total number in Table 4 because the CDP survey covers a shorter time period (from 2012 to 2021 of our sample period).

exhibit a significantly stronger increase in emissions-to-sales elasticity following a rise in interest rates compared to firm-years with such a target. The coefficient on the interaction term is approximately nine times larger for firms without carbon targets than for those with, and this difference is statistically significant.

Table 6 Panel C reports the results conditional on whether a firm-year adopts an intensity target. Unlike absolute targets, which focus on total emissions, intensity targets provide a more precise measure of a firm's carbon efficiency and decarbonization efforts (Aswani et al. 2024; Zhang 2025). We find that firm-years without an intensity target exhibit a significantly stronger increase in the emissions-to-sales elasticity following monetary tightening compared to firm-years with such a target, with the magnitude of the coefficient difference widening relative to those in Panel B.

Taken together, the results in Table 6 demonstrate the importance of management control tools in corporate responses to monetary shocks. Firms with stronger management control systems for decarbonization—such as climate-related performance incentives or explicit carbon targets—exhibit a weaker increase in carbon elasticity to sales following monetary tightening. This suggests that firms with a stronger commitment to environmental goals are less likely to scale back their decarbonization efforts in response to rising interest rates.

Analyses Conditional on Stakeholder Pressure

Prior research suggests that environmentally conscious stakeholders play a crucial role in shaping firms' environmental practices. To explore the role of stakeholder pressure, we conduct analysis conditional on whether a firm is headquartered in a democratic-leaning state or has high UN PRI ownership.

First, we partition firms based on the partisan leanings of the county where their corporate headquarters are located. We classify a county as a blue county if the percentage of voters supporting the Democratic candidate in the previous presidential election is above the sample median, and as a red county if it is below the sample median. Di Giuli and Kostovetsky (2014) find that firms headquartered in Democratic-leaning states have better corporate social responsibility performance. Since political views tend to cluster geographically (Porter 1998, 2000), firms headquartered in blue counties are more likely to operate in communities where stakeholders prioritize environmental concerns. As a result, these firms are subject to greater pressure to maintain their decarbonization efforts. We expect firms in blue counties to exhibit a weaker response to monetary policy shocks, as environmentally conscious stakeholders are more likely to hold them accountable for their environmental impact. Consistent with our expectation, Table 7 Panel A shows that firms headquartered in blue counties exhibit a significantly weaker response to monetary policy changes compared to firms in red counties.

Second, we partition our sample based on whether a firm's institutional ownership by UN PRI signatories is above or below the sample median in a given year. Prior studies suggest that institutional investors often pressure firms to reduce carbon emissions to minimize climate-related financial risks (Dyck et al. 2019; Krueger et al. 2020). UN PRI signatories commit to incorporating environmental, social, and governance (ESG) considerations into their investment decisions and actively engage with firms to improve their environmental performance. Consequently, we expect stronger stakeholder pressure among firms with higher UN PRI ownership.

We restrict this analysis to the post-2015 period because UN PRI signatories were less prominent in earlier years and their presence expanded significantly after 2015.[15] This timing aligns with the Paris Agreement, signed in December 2015, which heightened global awareness of climate change and corporate carbon emissions. Additionally, 2015 marked the establishment of the Task Force on Climate-related Financial Disclosures (TCFD), further reinforcing investor focus on climate-related risks. Table 7 Panel B shows that firms with above-median ownership by PRI signatories show a weaker response to monetary policy changes, reinforcing the idea that institutional investors play a disciplinary role in promoting environmentally responsible behavior. Overall, the findings in Table 7 suggest that firms facing stronger stakeholder pressure are less likely to scale back their decarbonization efforts in response to monetary tightening.

Analyses Conditional on Firm Financial Constraints

Firms' financial constraints can act as an amplification mechanism through which changes in interest rates affect not only the price but also the quantity of credit available to firms (Kiyotaki and Moore 1997). Much of the prior literature compares financially constrained to unconstrained firms and finds that financial constraints amplify monetary policy transmission (Ozdagli 2018; Chava and Hsu 2020; Ottonello and Winberry 2020; Cloyne et al. 2023).

The role of financial constraints in driving our findings is, ex ante, ambiguous. On one hand, financially constrained firms are more sensitive to changes in interest rates and may be more likely to forgo capital-intensive investments in clean technologies. On the other hand, firms may choose decarbonization projects due to political, social, and reputational considerations, rather than NPV maximization. When facing performance pressure, firms may decide to reduce these projects first. Additionally, constrained firms may already invest less in green investments

[15] See https://www.unpri.org/annual-report-2021/how-we-work/building-our-effectiveness/enhance-our-global-footprint

(Xu and Kim 2022).[16] As a result, even if these firms significantly cut green investments in response to monetary tightening, the overall impact on their emissions-to-sales elasticity may not necessarily be more pronounced than for unconstrained firms. Thus, the extent to which financial constraints shape the relationship between monetary policy and corporate carbon emissions remains an empirical question.

We use several proxies of firms' financial constraints (Farre-Mensa and Ljungqvist 2016). First, we use firm age and size. Young and small firms face greater frictions in accessing external financing due to less well-established relationships with financial markets, higher information asymmetry, and more uncertain returns. Thus, young firm age and small firm size (Gertler and Gilchrist 1994; Kashyap et al. 1994) are common proxies of financial constraints. Second, we use a composite financial constraint index proposed by Hadlock and Pierce (2010) that captures the combined effects of firm age and size.

To assess whether financially constrained firms exhibit stronger responses to monetary policy in their carbon reduction efforts, we re-estimate Equation (2), splitting the sample into more and less financially constrained firms. Table 8 presents the results. Panel A splits the sample by firm age, distinguishing young firms (below-median age in a given year) from older firms (above-median age). While the coefficient on the interaction term $\Delta FFR \times \Delta Log(Sales)$ is slightly higher for young firms than old firms, the difference in the coefficients on the interaction term $FFR\ Shock \times \Delta Log(Sales)$ is reversed. Moreover, the magnitudes of these differences are small

[16] It is possible that our sample consists of relatively large and older firms that are not financially constrained. Consistent with this view, additional analysis finds that the amount of decarbonization investment is significantly lower among the younger and smaller firms in our sample. For example, using firm size as a proxy, the mean and median levels of decarbonization investment for financially constrained firms are \$6,120,271 and \$682,863, respectively. In contrast, for non-financially constrained firms, the mean and median are significantly higher at \$39,900,000 and \$4,915,000. This confirms that financially constrained firms invest less in decarbonization projects—on average, their decarbonization investment amounts are approximately 15% (mean) and 13% (median) of those by non-financially constrained firms.

and statistically insignificant. Panel B reports the results conditional on firm size. We find that the coefficients on the interaction terms are slightly larger for smaller firms, but again, the differences across the subsamples are not statistically significant. Panel C presents the results conditional on the Hadlock and Pierce (2010) Index. We find no significant differences in the coefficients on the interaction terms between more and less financially constrained firms. Overall, the findings in Table 8 suggest that the financial constraint is unlikely to be the primary driver of our documented results.

VI. ROBUSTNESS CHECKS

We conduct a series of sensitivity tests to check the robustness of our primary results. Table 9 Panels A and B present the robustness checks for the results in Column (6) of Table 3 and Column (4) of Table 4, respectively. For brevity, we only tabulate the coefficients and statistical significance on the variable of interest.

Alternative Samples

To address the possibility that self-reported carbon emissions may lack credibility, we restrict our sample to firm-years with external verification/assurance for the reported carbon emissions. We obtain the data based on the question in the CDP survey: "Indicate the verification/assurance status that applies to your reported emissions." We find that our results remain qualitatively similar. In addition, since Table 1 Panel B shows that companies in the utility industry account for the highest number of observations in our sample, we rerun our analyses after excluding utility firms. The results remain robust.

Alternative Measures of FFR Shocks

We use two alternative measures of FFR shocks. The first follows Nakamura and Steinsson (2018), who construct monetary policy shocks using the first principal component of surprise changes in five futures contracts observed within a 30-minute window around scheduled FOMC announcements. These include the current-month federal funds futures, the expected rate after the next FOMC meeting, and three-month Eurodollar futures at two-, three-, and four-quarter horizons.

The second measure is a time-weighted monetary shock, following Ottonello and Winberry (2020). They aggregate high-frequency shocks into an annual variable by weighting each shock by the number of days remaining in the year. This approach captures firms' effective exposure to each shock, giving greater weight to shocks occurring earlier in the year when firms have more time to respond. We continue to find robust results.

Alternative Clustering Schemes

Lastly, we use an alternative clustering scheme at the year level to evaluate the significance of the regression coefficients. We find that our results remain robust.

VII. CONCLUSION

This study examines the effects of monetary policy on corporate carbon transition. We find that changes in monetary policies are positively associated with subsequent changes in firms' direct carbon emissions, as well as the elasticity of emissions to sales. Our analysis of project-level decarbonization investments finds that rising interest rates are negatively associated with the amount and payback period of these investments. These findings are consistent with the notion that monetary tightening diminishes the attractiveness of long-term, capital-intensive carbon-savings projects, leading to increased corporate carbon emissions. Furthermore, we find that the increase in the emissions-to-sales elasticity is less pronounced among firms with stronger carbon management control systems and those facing greater pressure from environmentally conscious stakeholders.

Overall, our findings suggest that monetary tightening hinders firms' green investments and decarbonization efforts, especially those with weak internal and external carbon governance mechanisms. We contribute to the literature by documenting how monetary policy shapes corporate green transition and highlighting the importance of management control and stakeholder scrutiny in the heterogeneity of firm-level responses.

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APPENDIX A
Excerpts from Cargill's Responses to CDP's 2022 Climate Change Questionnaire

Panel I: Carbon Reduction Initiatives

(C4.3b) Provide details on the initiatives implemented in the reporting year in the table below.

Initiative category & Initiative type

Energy efficiency in production processes	Waste heat recovery

Estimated annual CO2e savings (metric tonnes CO2e)
3147

Scope(s) or Scope 3 category(ies) where emissions savings occur
Scope 1

Voluntary/Mandatory
Voluntary

Annual monetary savings (unit currency – as specified in C0.4)
381311

Investment required (unit currency – as specified in C0.4)
601504

Payback period
1-3 years

Estimated lifetime of the initiative
6-10 years

Panel II: Provision of Monetary Incentives

(C1.3a) Provide further details on the incentives provided for the management of climate-related issues (do not include the names of individuals).

Entitled to incentive	Type of incentive	Activity incentivized	Comment
Corporate executive team	Monetary reward	Emissions reduction target	The Executive Team's compensation is based on a set of financial and performance metrics, and then adjusted based on select ESG targets

Panel III: Provision of Carbon Targets

(C4.1) Did you have an emissions target that was active in the reporting year?

Absolute target
Intensity target

APPENDIX B
Variable Definitions

Variable	Definition	Data Source
Variables of interest		
ΔLog(Scope1)	Change in the natural logarithm of firm Scope 1 carbon emissions	Trucost
ΔFFR	Annual change in the effective federal funds rate (in percentage points)	FRED
FFR Shock	Annual aggregate of monetary policy shocks identified by Jarociński and Karadi (2020)	Authors' website
Macroeconomic controls		
Log Industrial Production	Natural logarithm of Industrial Production Index	FRED
Log CPI	Natural logarithm of Consumer Price Index	FRED
Employment Rate	Employment-Population Ratio	FRED
ΔGDP	Percentage change in Real Gross Domestic Product	FRED
Excess Bond Premium	Excess bond premium from Gilchrist and Zakrajsek (2012)	Authors' website
D_Zero Bound	A variable indicating the zero-lower bound period (2009 to 2015)	
Firm-level controls		
Cashhold	Cash and short-term investments divided by total assets	Compustat
Cashflow	Operating income before depreciation divided by total assets	Compustat
ROA	Net income divided by total assets	Compustat
Leverage	Sum of short-term and long-term debt divided by total assets	Compustat
Size	Natural logarithm of total assets	Compustat
Tangibility	Property plant and equipment divided by total assets	Compustat
Tobin's Q	Sum of market value of equity and book value of debt divided by total assets	Compustat
ΔLog(Sale)	Change in the natural logarithm of firm sales	Compustat
Carbon initiative variables		
Log(Decarbonization Investment)	Natural logarithm of the investment (in $) for Scope 1 carbon emission reduction initiatives implemented during the reporting year	CDP
Log(Payback Period)	Natural logarithm of the payback period, calculated as the investment divided by annual monetary savings, for Scope 1 carbon emission reduction initiatives implemented during the reporting year	CDP
Log(Decarbonization /Total Investment)	Natural logarithm of the amount of the decarbonization investment scaled by the total amount of investment during the reporting year	CDP and Compustat
Partitioning variables		
Monetary incentives	Whether the firm provides monetary incentives for mangers to manage "climate change issues" (including carbon emissions)	CDP
Carbon targets	Whether the firm has any active Scope 1 emission targets, including both absolute targets and intensity targets	CDP
Intensity targets	Whether the firm has any active Scope 1 emission intensity targets	CDP

Blue counties	Counties where the percentage of voters supporting the Democratic candidate in the previous presidential election exceeds the sample median	MIT Election Data and Science Lab
UN PRI ownership	Institutional ownership by investors who are signatories to the United Nations Principles for Responsible Investment	Thomson/Refinitiv & PRI website
Firm age	The number of years the firm appears on Compustat with a non-missing stock price	Compustat
Hadlock and Pierce index	The size-age index developed by Hadlock and Pierce (2010)	Compustat

TABLE 1
Sample Selection and Distribution

Table 1 Panel A reports the sample selection procedure. Panel B presents the sample distribution by industry.

Panel A: Sample Selection

	N(firm-years)	N(firms)
U.S. firm-reported carbon emissions in Trucost, 2005-2021	6,923	1,119
Merge with Compustat	6,820	1,095
Exclude financial firms (SIC 6000-6999)	5,878	930
Exclude observations with missing required variables	4,582	603

Panel B: Sample Distribution by Industry

GICS Industry Classification	N(firm-years)	%	N(firms)	%
Automobiles & Components	87	1.9%	14	2.3%
Capital Goods	556	12.1%	71	11.8%
Consumer Durables & Apparel	97	2.1%	19	3.2%
Consumer Services	74	1.6%	12	2.0%
Diversified Financials	57	1.2%	9	1.5%
Energy	423	9.2%	64	10.6%
Food & Staples Retailing	59	1.3%	10	1.7%
Food, Beverage & Tobacco	297	6.5%	34	5.6%
Health Care Equipment & Services	37	0.8%	5	0.8%
Household & Personal Products	86	1.9%	8	1.3%
Materials	521	11.4%	72	11.9%
Media & Entertainment	39	0.9%	7	1.2%
Pharmaceuticals, Biotech. & Life Sciences	180	3.9%	21	3.5%
Retailing	216	4.7%	31	5.1%
Semiconductors & Semiconductor Equip.	192	4.2%	29	4.8%
Software & Services	229	5.0%	33	5.5%
Technology Hardware & Equipment	357	7.8%	44	7.3%
Telecommunication Services	95	2.1%	15	2.5%
Transportation	227	5.0%	29	4.8%
Utilities	753	16.4%	76	12.6%
Total	**4,582**	100.0%	**603**	100.0%

TABLE 2
Summary Statistics

Panel A reports summary statistics for the primary variables used in the analyses. Panel B reports Pearson correlation coefficients. A correlation coefficient in bold indicates that the correlation is significant at better than the 10% two-tailed level. Appendix B provides variable definitions.

Panel A: Summary Statistics (N=4,582 firm-years)

Variables	Mean	Std. Dev.	Q1	Median	Q3
ΔLog(Scope1)	-0.011	0.281	-0.105	-0.005	0.086
ΔFFR	-0.149	1.104	-0.720	0.030	0.300
FFR Shock	-0.069	0.113	-0.133	-0.070	0.007
Log Industrial Production	4.592	0.041	4.577	4.600	4.624
Log CPI	4.602	0.078	4.563	4.608	4.670
Employment Rate	59.720	1.508	58.600	59.600	60.600
ΔGDP	1.838	1.533	1.552	2.182	2.994
Excess Bond Premium	0.038	0.437	-0.150	-0.102	0.011
D_Zeo Bound	0.396	0.489	0.000	0.000	1.000
Cashhold	0.111	0.115	0.026	0.074	0.156
Cashflow	0.143	0.080	0.092	0.131	0.183
ROA	0.052	0.071	0.023	0.051	0.089
Leverage	0.322	0.156	0.214	0.313	0.416
Size	9.623	1.211	8.764	9.594	10.470
Tangibility	0.356	0.258	0.128	0.275	0.587
Tobin's Q	1.989	1.205	1.228	1.569	2.285
ΔLog(Sale)	0.032	0.175	-0.030	0.036	0.101

TABLE 2, CONTINUED

Panel B: Correlation Coefficients (N=4,582 firm-years)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1)	1															
(2)	**0.05**	1														
(3)	**0.04**	**0.87**	1													
(4)	-0.02	**0.31**	**0.35**	1												
(5)	**0.03**	**-0.06**	-0.01	**0.50**	1											
(6)	**-0.08**	**0.19**	**0.06**	**0.39**	**-0.35**	1										
(7)	**-0.03**	**0.72**	**0.68**	**0.54**	0.01	**0.40**	1									
(8)	-0.03	**-0.64**	**-0.75**	**-0.63**	**-0.21**	**-0.15**	**-0.70**	1								
(9)	**-0.03**	**-0.20**	0.02	**-0.48**	**-0.43**	**-0.43**	**-0.15**	**0.30**	1							
(10)	**-0.05**	**-0.03**	-0.03	**-0.03**	0.01	**-0.06**	-0.01	**0.04**	**0.06**	1						
(11)	**0.10**	**0.06**	**0.04**	**-0.05**	**-0.06**	**-0.04**	-0.01	-0.01	0.03	**0.22**	1					
(12)	**0.06**	**0.05**	**0.03**	**-0.06**	**-0.05**	**-0.06**	**-0.03**	-0.01	**0.03**	**0.23**	**0.76**	1				
(13)	0.01	-0.00	-0.03	**0.18**	**0.26**	0.01	**0.04**	**-0.06**	**-0.21**	**-0.28**	**-0.13**	**-0.23**	1			
(14)	0.02	0.02	0.01	-0.02	**-0.06**	0.02	-0.00	0.00	0.01	**-0.12**	-0.02	**0.04**	-0.01	1		
(15)	0.02	0.00	-0.01	0.01	**-0.03**	**0.07**	0.00	-0.00	**-0.04**	**-0.46**	**-0.19**	**-0.26**	**0.21**	**0.10**	1	
(16)	0.02	0.02	0.02	**0.11**	**0.17**	**-0.06**	0.01	**-0.07**	**-0.10**	**0.39**	**0.59**	**0.54**	**-0.04**	**-0.10**	**-0.34**	1
(17)	**0.27**	**0.11**	**0.08**	**-0.10**	0.02	**-0.16**	**-0.08**	**-0.11**	**-0.10**	0.02	**0.39**	**0.31**	**-0.08**	**0.03**	**-0.05**	**0.16**

Variables:

(1) ΔLog(Scope1)	(7) ΔGDP	(13) Leverage
(2) ΔFFR	(8) Excess Bond Premium	(14) Size
(3) FFR Shock	(9) D_Zero Bound	(15) Tangibility
(4) Log Industrial Production	(10) Cashhold	(16) Tobin's Q
(5) Log CPI	(11) Cashflow	(17) ΔLog(Sale)
(6) Employment Rate	(12) ROA	

TABLE 3
Effect of Monetary Policy on Corporate Carbon Emissions

Table 3 presents the results of the effect of monetary policy on corporate Scope 1 emissions. *ΔLog(Scope1)* is the change in the log of a firm's Scope 1 carbon emissions. *ΔFFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. See Appendix B for definitions of other variables. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Dep. Var. =	**ΔLog(Scope1)**					
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
ΔFFR	**0.022****	**0.018****	**0.015*****			
	(2.71)	**(2.55)**	**(3.20)**			
FFR Shock				**0.196****	**0.160****	**0.138****
				(2.27)	**(2.21)**	**(2.79)**
Log Industrial Production	0.277	0.297	0.529**	0.440	0.428	0.641**
	(0.69)	(0.89)	(2.53)	(1.10)	(1.30)	(2.89)
Log CPI	-0.380	-0.479**	-0.380***	-0.566**	-0.625***	-0.503***
	(-1.42)	(-2.15)	(-3.08)	(-2.24)	(-3.05)	(-4.15)
Employment Rate	-0.027*	-0.028**	-0.018***	-0.037**	-0.035***	-0.025***
	(-1.92)	(-2.43)	(-3.35)	(-2.70)	(-3.29)	(-4.73)
ΔGDP	-0.015	-0.010	-0.005	-0.007	-0.003	0.001
	(-1.74)	(-1.38)	(-0.98)	(-0.89)	(-0.57)	(0.14)
Excess Bond Premium	-0.006	0.005	0.036*	0.025	0.030	0.058**
	(-0.18)	(0.17)	(1.91)	(0.67)	(1.00)	(2.82)
D_Zero Bound	-0.054	-0.046	-0.015	-0.089**	-0.074***	-0.038***
	(-1.59)	(-1.69)	(-1.10)	(-2.86)	(-3.02)	(-2.99)
Cashhold		-0.398***	-0.305***		-0.400***	-0.306***
		(-3.39)	(-3.12)		(-3.42)	(-3.14)
Cashflow		0.707***	0.079		0.710***	0.08
		(5.65)	(0.69)		(5.67)	(0.70)
ROA		-0.147	-0.110		-0.141	-0.105
		(-1.08)	(-0.90)		(-1.05)	(-0.87)
Leverage		0.118*	0.101		0.120*	0.103
		(1.90)	(1.65)		(1.94)	(1.69)
Size		0.067**	0.043*		0.067**	0.043*
		(2.79)	(1.81)		(2.77)	(1.79)
Tangibility		-0.129	-0.027		-0.128	-0.025
		(-1.31)	(-0.23)		(-1.30)	(-0.22)
Tobin's Q		-0.015**	-0.007		-0.015**	-0.007
		(-2.15)	(-1.03)		(-2.12)	(-1.02)
ΔLog(Sale)			0.367***			0.367***
			(8.33)			(8.33)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
N(firm-years)	4,582	4,582	4,582	4,582	4,582	4,582
Adj. R-squared	0.170	0.191	0.217	0.170	0.190	0.216

TABLE 4

Effect of Monetary Policy on the Elasticity of Carbon Emissions to Sales

Table 4 presents the results of the effect of monetary policy on the elasticity of firms' Scope 1 carbon emissions to sales. Δ*Log(Scope1)* is the change in the log of a firm's Scope 1 carbon emissions. Δ*FFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. Δ*Log(Sale)* is the change in the log of a firm's sales. See Appendix B for definitions of other variables. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
ΔLog(Sale)	0.416***	0.394***	0.447***	0.423***
	(9.99)	(8.65)	(10.13)	(8.77)
ΔFFR × ΔLog(Sale)	**0.077*** **	**0.071** **		
	(2.95)	**(2.72)**		
FFR Shock × ΔLog(Sale)			**0.687** **	**0.647** **
			(2.92)	**(2.69)**
Cashhold		-0.286**		-0.287**
		(-2.82)		(-2.88)
Cashflow		0.085		0.087
		(0.73)		(0.75)
ROA		-0.109		-0.111
		(-0.87)		(-0.89)
Leverage		0.084		0.086
		(1.37)		(1.42)
Size		0.043*		0.044*
		(1.88)		(1.87)
Tangibility		-0.011		-0.012
		(-0.10)		(-0.11)
Tobin's Q		-0.007		-0.008
		(-1.11)		(-1.10)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N(firm-years)	4,582	4,582	4,582	4,582
Adj. R-squared	0.214	0.220	0.214	0.220

TABLE 5
Monetary Policy and Decarbonization Investments

Table 5 presents the results on the effect of monetary policy on firms' decarbonization investments implemented each year. Panel A summarizes the decarbonization projects by category. Panel B provides summary descriptive statistics. Panel C presents the regression results. ΔFFR is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. See Appendix B for definitions of other variables. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Description of Decarbonization Investments

Category	CDP Activity Type	N (projects)	%	%(sum)
Building efficiency	Energy efficiency: Building services	1,252	31.6%	
	Energy efficiency in buildings	262	6.6%	
	Energy efficiency: Building fabric	216	5.5%	43.7%
Process efficiency	Energy efficiency: Processes	865	21.9%	
	Energy efficiency in production processes	398	10.1%	
	Process emissions reductions	153	3.9%	
	Non-energy industrial process emissions reductions	18	0.5%	36.2%
Transportation	Transportation	44	1.1%	
	Transportation: fleet	141	3.6%	
	Transportation: use	31	0.8%	5.5%
Low carbon energy	Low-carbon energy installation	147	3.7%	
	Low-carbon energy generation	27	0.7%	
	Low-carbon energy consumption	25	0.6%	
	Low-carbon energy purchase	17	0.4%	5.5%
Behavioral change	Behavioral change	89	2.2%	
	Company policy or behavioral change	15	0.4%	2.6%
Waste	Fugitive emissions reductions	89	2.2%	
	Waste reduction and material circularity	8	0.2%	
	Waste recovery	3	0.1%	2.5%
Product design	Product design	26	0.7%	0.7%
Other	Other	129	3.3%	
	Green project finance	1	0.0%	3.3%
Total		3,956	100.0%	100.0%

TABLE 5, CONTINUED

Panel B: Descriptive Statistics of the Carbon Reduction Initiatives (N=3,956 projects)

	Mean	Std. Dev.	Q1	Median	Q3
Log(Decarbonization Investment)	12.750	3.055	11.120	12.820	14.630
Log(Payback Period)	0.704	2.135	0.0925	0.938	1.609
Log(Decarbonization/Total Investment)	-9.265	3.093	-10.960	-9.227	-7.387

Panel C: Regression Results

Dep. Var. =	Log(Decarbonization Investment)		Log(Payback Period)		Log(Decarbonization /Total Investment)	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔFFR	**-0.500*** **		**-0.358*** **		**-0.510*** **	
	(-7.81)		**(-6.68)**		**(-7.22)**	
FFR Shock		**-3.243*** **		**-1.746** **		**-3.651*** **
		(-4.91)		**(-3.13)**		**(-5.83)**
Log Ind. Prod.	12.935***	7.184***	10.060***	5.714***	11.807***	6.071**
	(5.68)	(3.59)	(4.74)	(3.33)	(5.10)	(3.06)
Log CPI	-10.091***	0.516	-7.729***	0.338	-8.633***	1.914
	(-6.36)	(0.35)	(-5.77)	(0.27)	(-5.21)	(1.33)
Employment Rate	-0.191**	-0.044	-0.004	0.049	-0.167*	0.013
	(-2.35)	(-0.84)	(-0.07)	(0.92)	(-1.96)	(0.25)
ΔGDP	0.071	0.042	-0.080**	-0.064	0.073	0.021
	(1.44)	(1.22)	(-2.55)	(-1.45)	(1.35)	(0.69)
Excess Bond Premium	-0.248	0.815	-0.619**	0.226	-0.250	0.786
	(-0.48)	(1.75)	(-2.34)	(0.53)	(-0.46)	(1.76)
Cashhold	1.763	1.498	0.810	0.657	2.296*	2.005*
	(1.64)	(1.46)	(0.97)	(0.79)	(2.11)	(1.95)
Cashflow	1.751	1.974	-1.202	-1.089	2.004	2.260
	(1.04)	(1.12)	(-0.99)	(-0.87)	(1.34)	(1.44)
ROA	-0.602	-0.449	0.157	0.225	-1.195	-1.014
	(-0.44)	(-0.33)	(0.14)	(0.20)	(-0.85)	(-0.72)
Leverage	-0.794	-0.775	0.305	0.322	-0.679	-0.661
	(-1.28)	(-1.25)	(0.68)	(0.73)	(-1.18)	(-1.15)
Size	0.131	0.094	-0.083	-0.105	-0.587*	-0.627**
	(0.47)	(0.33)	(-0.48)	(-0.60)	(-2.22)	(-2.31)
Tangibility	2.806	2.714	0.670	0.613	0.895	0.796
	(1.68)	(1.61)	(0.47)	(0.42)	(0.55)	(0.48)
Tobin's Q	-0.240	-0.279*	-0.066	-0.086	-0.306**	-0.350**
	(-1.77)	(-1.99)	(-0.70)	(-0.89)	(-2.26)	(-2.49)
ΔLog(Sale)	0.432	0.431	0.578*	0.548*	0.267	0.284
	(1.08)	(1.13)	(1.85)	(1.95)	(0.66)	(0.74)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
N(projects)	3,956	3,956	3,956	3,956	3,956	3,956
Adj. R-squared	0.567	0.567	0.475	0.475	0.579	0.579

TABLE 6
Analyses Conditional on Carbon Management Control Systems

Table 6 presents the results of the effect of monetary policy on the emissions-to-sales elasticity conditional on carbon managerial control systems. Panels A, B, and C partition the samples by whether a firm-year provides monetary incentives for climate-related goals, has active carbon targets, and has active carbon intensity targets, respectively. Δ*Log(Scope1)* is the change in the log of a firm's Scope 1 emissions. Δ*FFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. Δ*Log(Sale)* is the change in the log of a firm's sales. Controls include *Cashhold*, *Cashflow*, *ROA*, *Leverage*, *Size*, *Tangibility*, and *Tobin's Q*. See Appendix B for definitions of other variables. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Analysis Conditional on Monetary Incentives

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**with monetary incentives**	**without monetary incent.**	**with monetary incentives**	**without monetary incent.**
ΔLog(Sale)	0.497***	0.342***	0.516***	0.450***
	(8.37)	(4.09)	(5.87)	(4.99)
ΔFFR × ΔLog(Sale)	**0.009**	**0.245*** **		
	(0.11)	**(4.11)**		
FFR Shock × ΔLog(Sale)			**0.308**	**2.652*** **
			(0.37)	**(3.86)**
Diff in Coeff.	***-0.236*** **		***-2.344****	
	(-2.31)		***(-2.15)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N (firm-years)	1,952	1,421	1,952	1,421
Adj. R-squared	0.273	0.301	0.273	0.303

Panel B: Analysis Conditional on Carbon Targets

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**with carbon targets**	**without carbon targets**	**with carbon targets**	**without carbon targets**
ΔLog(Sale)	0.520***	0.334***	0.532***	0.426***
	(4.95)	(4.02)	(4.08)	(5.11)
ΔFFR × ΔLog(Sale)	**0.023**	**0.189*** **		
	(0.27)	**(3.76)**		
FFR Shock × ΔLog(Sale)			**0.246**	**2.290*** **
			(0.32)	**(3.78)**
Diff in Coeff.	***-0.167*** **		***-2.044****	
	(-2.25)		***(-2.21)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	1,606	1,725	1,606	1,725
Adj. R-squared	0.340	0.258	0.253	0.260

TABLE 6, CONTINUED

Panel C: Analysis Conditional on Intensity Targets

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**with intensity targets**	**without intensity targets**	**with intensity targets**	**without intensity targets**
ΔLog(Sale)	0.444***	0.400***	0.314**	0.487***
	(4.87)	(5.94)	(2.55)	(7.61)
ΔFFR × ΔLog(Sale)	**-0.177**	**0.193*** **		
	(-1.29)	**(4.98)**		
FFR Shock × ΔLog(Sale)			**-2.389***	**2.150*** **
			(-2.02)	**(5.08)**
Diff in Coeff.	***-0.370*** **		***-4.538****	
	(-2.99)		***(-4.17)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	832	2,568	832	2,568
Adj. R-squared	0.346	0.246	0.348	0.247

TABLE 7
Analyses Conditional on Stakeholder Pressure

Table 7 presents the results of the effect of monetary policy on the emissions-to-sales elasticity conditional on stakeholder pressure. Panels A and B partition the samples by the political leaning of the county where the firm is headquartered (blue vs. red counties) and by institutional ownership of UN PRI signatories (above vs below sample median), respectively. Panel A uses the full sample period, while Panel B focuses on data since 2015, when UN PRI became more widely adopted. Δ*Log(Scope1)* is the change in the log of a firm's Scope 1 emissions. Δ*FFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. Δ*Log(Sale)* is the change in the log of a firm's sales. Controls include *Cashhold*, *Cashflow*, *ROA*, *Leverage*, *Size*, *Tangibility*, and *Tobin's Q*. See Appendix B for definitions of other variables. t-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Analysis Conditional on Democratic-Leaning Counties, All Years

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**Blue counties**	**Red counties**	**Blue counties**	**Red counties**
ΔLog(Sale)	0.407***	0.398***	0.411***	0.451***
	(6.77)	(7.85)	(5.95)	(8.97)
ΔFFR × ΔLog(Sale)	**0.026**	**0.108****		
	(1.01)	(2.60)		
FFR Shock × ΔLog(Sale)			**0.174**	**1.042****
			(0.87)	(2.64)
Diff in Coeff.	***-0.082****		***-0.869*****	
	(-1.91)		***(-2.11)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N (firm-years)	2,208	2,263	2,208	2,263
Adj. R-squared	0.234	0.227	0.234	0.226

Panel B: Analysis Conditional on UN PRI Ownership, 2015-2021

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**High PRI ownership**	**Low PRI ownership**	**High PRI ownership**	**Low PRI ownership**
ΔLog(Sale)	0.286***	0.571***	0.273*	0.704***
	(4.20)	(7.38)	(2.38)	(8.18)
ΔFFR × ΔLog(Sale)	**-0.058**	**0.136****		
	(-0.85)	(2.76)		
FFR Shock × ΔLog(Sale)			**-0.360**	**2.078*****
			(-0.37)	(4.14)
Diff in Coeff.	***-0.194*****		***-2.438*****	
	(-2.58)		***(-2.46)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	1,166	1,174	1,166	1,174
Adj. R-squared	0.283	0.372	0.283	0.373

TABLE 8
Analyses Conditional on Financial Constraints

Table 8 presents the results of the effect of monetary policy on the emissions-to-sales elasticity conditional on firm financial constraints. Panels A, B, and C partition the samples by firm age, size, and the Hadlock and Pierce (2010) index, respectively. Δ*Log(Scope1)* is the change in the log of a firm's Scope 1 emissions. Δ*FFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. Controls include *Cashhold*, *Cashflow*, *ROA*, *Leverage*, *Size*, *Tangibility*, and *Tobin's Q*. See Appendix B for variable definitions. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Analysis Conditional on Firm Age

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**Young**	**Old**	**Young**	**Old**
ΔLog(Sale)	0.335*** (5.66)	0.495*** (5.87)	0.359*** (5.15)	0.537*** (6.38)
ΔFFR × ΔLog(Sale)	**0.082**** **(2.68)**	**0.065*** **(1.99)**		
FFR Shock × ΔLog(Sale)			**0.640**** **(2.17)**	**0.757**** **(2.47)**
Diff in Coeff.	***0.062*** ***(0.45)***		***-0.117*** ***(-0.36)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	2,328	2,216	2,328	2,216
Adj. R-squared	0.241	0.243	0.241	0.244

Panel B: Analysis Conditional on Firm Size

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**Small**	**Large**	**Small**	**Large**
ΔLog(Sale)	0.300*** (3.30)	0.450*** (6.30)	0.336*** (3.33)	0.471*** (6.06)
ΔFFR × ΔLog(Sale)	**0.083**** **(2.24)**	**0.054**** **(2.87)**		
FFR Shock × ΔLog(Sale)			**0.734*** **(1.76)**	**0.509*** **(2.11)**
Diff in Coeff.	***0.029*** ***(0.74)***		***0.2254*** ***(0.45)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	2,220	2,300	2,220	2,300
Adj. R-squared	0.234	0.262	0.233	0.261

TABLE 8, CONTINUED

Panel C: Analysis Conditional on Hadlock and Pierce (2010) Index

Dep. Var. =	**ΔLog(Scope1)**			
	(1)	**(2)**	**(3)**	**(4)**
Partition =	**High Index**	**Low Index**	**High Index**	**Low Index**
ΔLog(Sale)	0.330***	0.486***	0.353***	0.527***
	(5.51)	(6.22)	(5.01)	(6.66)
ΔFFR × ΔLog(Sale)	**0.078****	**0.058***		
	(2.84)	**(1.93)**		
FFR Shock × ΔLog(Sale)			**0.625****	**0.710****
			(2.32)	**(2.48)**
Diff in Coeff.	***0.020***		***-0.085***	
	(0.55)		***(-0.28)***	
Controls	Yes	Yes	Yes	Yes
Fixed effects	Firm FE, Year FE			
N(firm-years)	2,237	2,310	2,237	2,310
Adj. R-squared	0.247	0.231	0.246	0.231

TABLE 9
Robustness Tests

Table 9 presents the results of the robustness checks of the association between monetary policy and corporate green transition. Panel A reports the tests for carbon emissions and Panel B for the elasticity of carbon emissions to sales. Δ*Log(Scope1)* is the change in the log of a firm's Scope 1 carbon emissions. Δ*FFR* is the annual change in the federal funds rate. *FFR Shock* is the annual funds-rate shock. Δ*Log(Sale)* is the change in the log of a firm's sales. See Appendix B for variable definitions. *t*-statistics are reported in parentheses, based on standard errors clustered at both the firm and year levels. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Robustness Checks of the Results in Column (6) of Table 3

Dep var. = ΔLog(Scope1)	**Coefficient on *FFR Shock***	**(t-stats.)**	**N(firm-years)**	**Adj. R^2**
Alternative sample				
1. Restricting to firms with carbon assurance	0.134**	(2.39)	2,300	0.256
2. Excluding utility firms	0.117**	(2.36)	3,859	0.218
Alternative measure of FFR shocks				
1. Nakamura and Steinsson (2018) measure	0.287***	(3.54)	4,582	0.216
2. Time-weighted shocks	0.273***	(2.95)	4,582	0.216
Alternative clustering schemes				
Clustering by year	0.138***	(3.41)	4,582	0.216

Panel B: Robustness Checks of the Results in Column (4) of Table 4

Dep var. = ΔLog(Scope1)	**Coefficient on *FFR Shock* × Δ*Log(Sale)***	**(t-stats.)**	**N(firm-years)**	**Adj. R^2**
Alternative sample				
1. Restricting to firms with carbon assurance	0.625*	(1.99)	2,300	0.264
2. Excluding utility firms	0.629**	(2.30)	3,859	0.221
Alternative measure of FFR shocks				
1. Nakamura and Steinsson (2018) measure	0.661**	(2.48)	4,582	0.219
2. Time-weighted shocks	1.196**	(2.20)	4,582	0.216
Alternative clustering schemes				
Clustering by year	0.647**	(2.57)	4,582	0.219